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# Import libraries
import pandas as pd
import numpy as np
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, f1_score, classification_report

# Load dataset
data = load_breast_cancer()
X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target)

# For demonstration, simulate issue priority:
# Let's say: malignant (1) = 'high' priority, benign (0) = 'low' priority
# Convert to numeric labels: high=2, medium=1, low=0 for a three-class example
# Since dataset is binary, we assign 'medium' randomly for some samples

np.random.seed(42)
priority = y.copy()
priority = priority.replace({0: 0, 1: 2}) # 0=low, 2=high
# Randomly assign 20% of samples as medium (1)
medium_indices = np.random.choice(priority.index, size=int(0.2*len(priority)), replace=False)
priority.loc[medium_indices] = 1

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(X, priority, test_size=0.2, random_state=42)

# Initialize Random Forest model
rf = RandomForestClassifier(random_state=42)

# Train model
rf.fit(X_train, y_train)

# Predict
y_pred = rf.predict(X_test)

# Evaluate
accuracy = accuracy_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred, average='weighted')

print("Accu

```

Task 3 Summary: Predictive Analytics for Resource Allocation

In this task, the Breast Cancer dataset was used to simulate predicting issue priority levels in software projects. The original binary labels (benign/malignant) were adapted to represent three priority classes: low, medium, and high. A Random Forest classifier was trained on the features to predict these priorities. The dataset was split into training and testing subsets to evaluate model performance fairly.

The model achieved a notable accuracy and weighted F1-score, demonstrating its ability to correctly classify most samples. The weighted F1-score is particularly useful here as it accounts for the imbalance between priority classes, providing a balanced measure of precision and recall.

This predictive approach illustrates how machine learning models can support resource allocation by forecasting issue priorities, enabling teams to focus efforts efficiently. Using AI in this way enhances decision-making in software engineering, ensuring critical issues receive timely attention. Future work could improve the model by incorporating more real-world data and addressing potential biases in the dataset.
